

LESSON PLAN	COURSE: MHF4U
Date: Lesson #11	Unit of Study: Chapter #6 Exponential and Logarithmic Functions
Materials / Resources: - Advanced Function 12 Study Guide - Chapter #6 & 7 Pre-assessment Quiz - Chapter #6 & 7 Formula, Aid and Homework Sheet(s)	Lesson Focus - The Exponential Function and Its Inverse - Logarithmic Functions
Prior Knowledge: - Characteristics of exponential functions - Knowledge of simple exponential equations and how to solve them	
Overall Expectations: 1. Demonstrate an understanding of the relationship between exponential expressions and logarithmic expressions, evaluate logarithms, and apply the laws of logarithms to simplify numeric expressions 2. Identify and describe some key features of the graphs of logarithmic functions, make connections among the numeric, graphical, and algebraic representations of logarithmic functions, and solve related problems graphically	
Specific Expectations: 1.1 Recognize the logarithm of a number to a given base as the exponent to which the base must be raised to get the number, recognize the operation of finding the logarithm to be the inverse operation (i.e., the undoing or reversing) of exponentiation, and evaluate simple logarithmic expressions 1.2 Determine, with technology, the approximate logarithm of a number to any base, including base 10 (e.g., by reasoning that 29 is between 3 and 4 and use systematic trial to determine that 29 is approximate 3.07) 1.3 Make connections between related logarithmic and exponential equations (e.g., $125 = 5^3$ can also be expressed as $5^3 = 125$) and solve simple exponential equations by rewriting them in logarithmic form (e.g., solving $3^x = 10$ by rewriting the equation as $10 = x$) 2.1 determine, through investigation with technology (e.g., graphing calculator, spreadsheet) and without technology, key features (i.e., vertical and horizontal asymptotes, domain and range, intercepts, increasing/decreasing behaviour) of the graphs of logarithmic functions of the form $f(x) = \log_b(x)$, and make connections between the algebraic and graphical representations of these logarithmic functions 2.2 recognize the relationship between an exponential function and the corresponding logarithmic function to be that of a function and its inverse, deduce that the graph of a logarithmic function is the reflection of the graph of the corresponding exponential function in the line $y = x$, and verify the deduction using technology	

Learning Goals:

- Students will be able to identify the characteristics of an exponential function
- Students will be able to identify the characteristics of the inverse of an exponential function
- Students will understand that logarithmic functions are inverses of unique exponential functions
- Students will learn how to write an exponential function in logarithmic form and vice versa
- Students will know how to estimate and evaluate a logarithm

Success Criteria:

1. By the end of this lesson I will be able to identify the nature of the rate of change and key features of an exponential function
2. By the end of this lesson I will be able to identify the shape of the graph and key features of the inverse of an exponential function
3. By the end of this lesson I will be able to identify logarithmic function as the inverse of a corresponding exponential function
4. By the end of this lesson I will be able to write an exponential equation in logarithmic form and vice versa
5. By the end of this lesson I will be able to estimate and evaluate a logarithm

Assessment Strategies:

Diagnostic: chapter 6 & 7 pre-assessment quiz

Formative: homework for chapters 6.1 & 6.2

Summative: quiz #5, test #4

Time:

Part 1

Introduction:

Students will be given a pre-assessment quiz to determine their knowledge standings on exponential equations, and then proceed from their answers.

Time:

Part 2

Lesson:

Chapter 6.1 The Exponential Function and Its Inverse

Refresh the student's memory in exponential functions and their characteristics, identify the function $y = b^x$ for $b > 1$ and $0 < b < 1$. Revisit on what it means for a function to have an inverse, and the main idea that functions are reflected in the xy-plane through the line $y = x$. Introduce the inverse of exponential functions and their characteristics. Take notice on how the characteristics of the exponential functions relates to the characteristics of the inverse exponential function.

Chapter 6.2 Logarithm

Define the inverse of the exponential function as a logarithmic function. Identify the characteristics of logarithmic functions (e.g., $y = \log \log x$ has a vertical asymptote at $x = 0$ and intersects the x-axis at $x = 1$). Show how to write an exponential equation

	in logarithmic form and how to write a logarithmic function in exponential form. Name the restrictions with exponential and logarithmic functions (e.g., the variable b in the functions $y = b^x \leftrightarrow y = x$ must be greater than zero, excluding 1 and x must be greater than zero for $y = b^x$ and y must be greater than zero for $y = x$.
Time: Part 3	Wrap-up: Ask students to demonstrate their acquired knowledge by applying them to example problems and then answer any unresolved questions.
Notes, reminders & homework: <u>Reminder:</u> - Quiz #5 (to be written before / after lesson #13) <u>Homework:</u> Chapter 6.1: #1, 3, 7, 8 Chapter 6.2: # 1 – 11, 15	
Lesson Reflection:	

